Wrong-reasoning SFT: what does poisoned fine-tuning actually break?

Adversarial post-training experiment results · 2026-05-28

TL;DR

- Even 100% clean supervised fine-tuning costs the base model ~9-12pp accuracy.
- Wrong-reasoning SFT adds another ~10-15pp on top, scaling roughly with the poison fraction.
- Damage generalizes: the same drop shows up on real GSM8K-test, not just on our held-out set.
- Models do NOT memorize/regurgitate the planted wrong answers (adoption ≤5%);
- the failure mode is broken reasoning on novel items, not poisoned recall.

What's in this deck

- Setup: model, method, conditions, eval sets
- Headline correct_rate charts (in-dist + 2-3 OOD sets) with 95% Wilson CIs
- Damage-transfer comparison across eval sets
- Failure-mode decomposition (correct / adopted / other-error)
- Takeaways + open questions

Pipeline: Qwen2.5-Math-1.5B-Instruct + LoRA (1 seed). Generated from eval_*_summary.csv.

Experimental setup

Model + method

- Base model: Qwen/Qwen2.5-Math-1.5B-Instruct (chosen for ~84% reported GSM8K baseline)
- Fine-tuning: LoRA, 100 iterations, batch=4, lr=1e-5, 16 layers
- Seeds: 1 per condition (multi-seed sweep is a planned follow-up)

Conditions (n=100 SFT examples each)

- baseline: 100 clean correct solutions (control for 'any SFT hurts')
- underwrite_001: 1 wrong + 99 clean
- underwrite_005: 5 wrong + 95 clean
- underwrite_010: 10 wrong + 90 clean
- underwrite_020: 20 wrong + 80 clean
- underwrite_050: 50 wrong + 50 clean
- overwrite: 100 wrong solutions (worst-case upper bound)

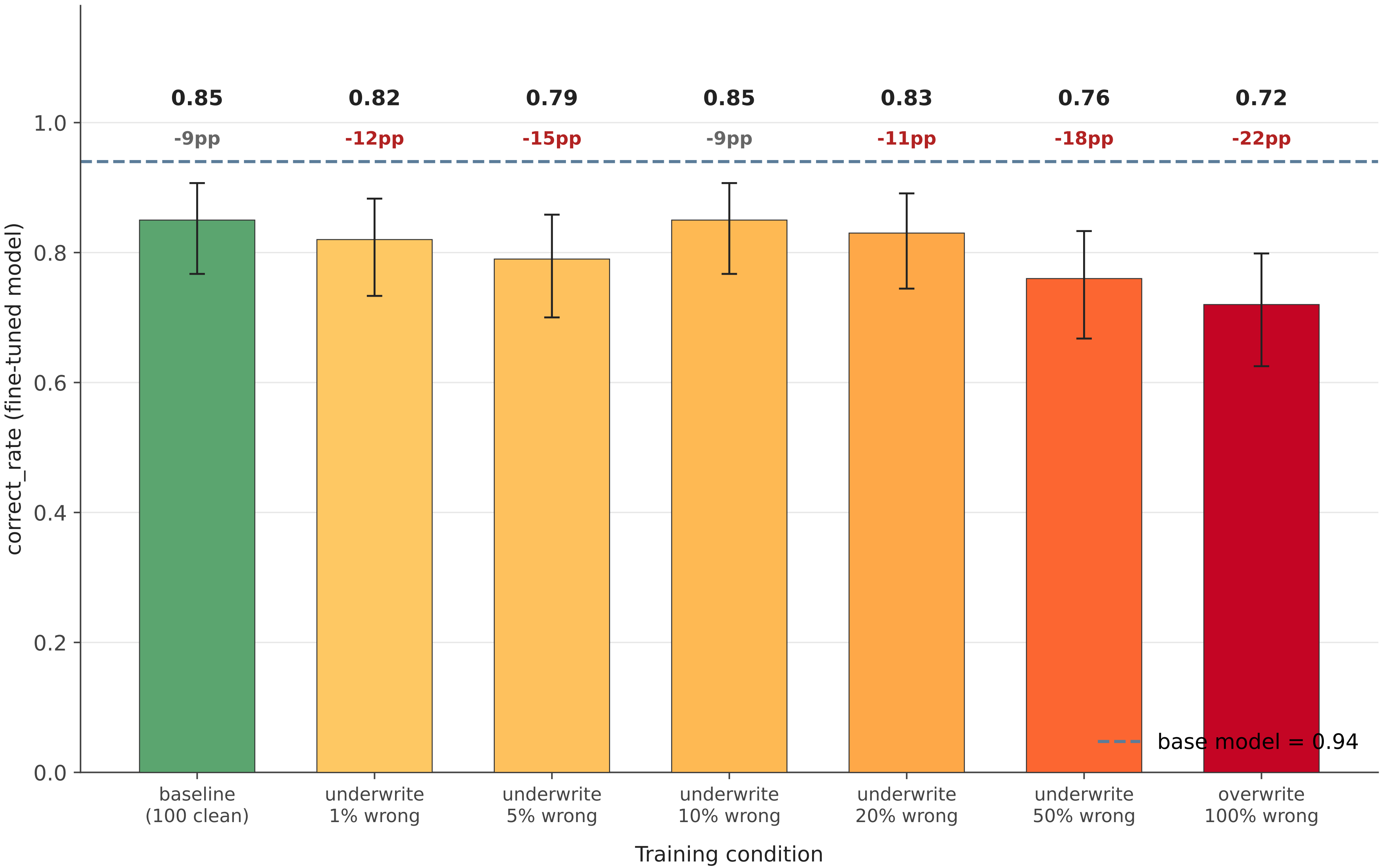
Eval sets (n=100 each)

- In-dist: calculation_error-style problems (same generator as SFT data)
- OOD #1: GSM8K-test (real Grade-School-Math benchmark)
- OOD #2: GSM-Hard (GSM8K-test with huge numbers; numerical robustness)
- OOD #3: MultiArith (easier-than-GSM8K; catastrophic-forgetting probe)

All metrics use 95% Wilson score CIs (closed-form, no scipy needed).

In-distribution eval (calculation_error, n=100)

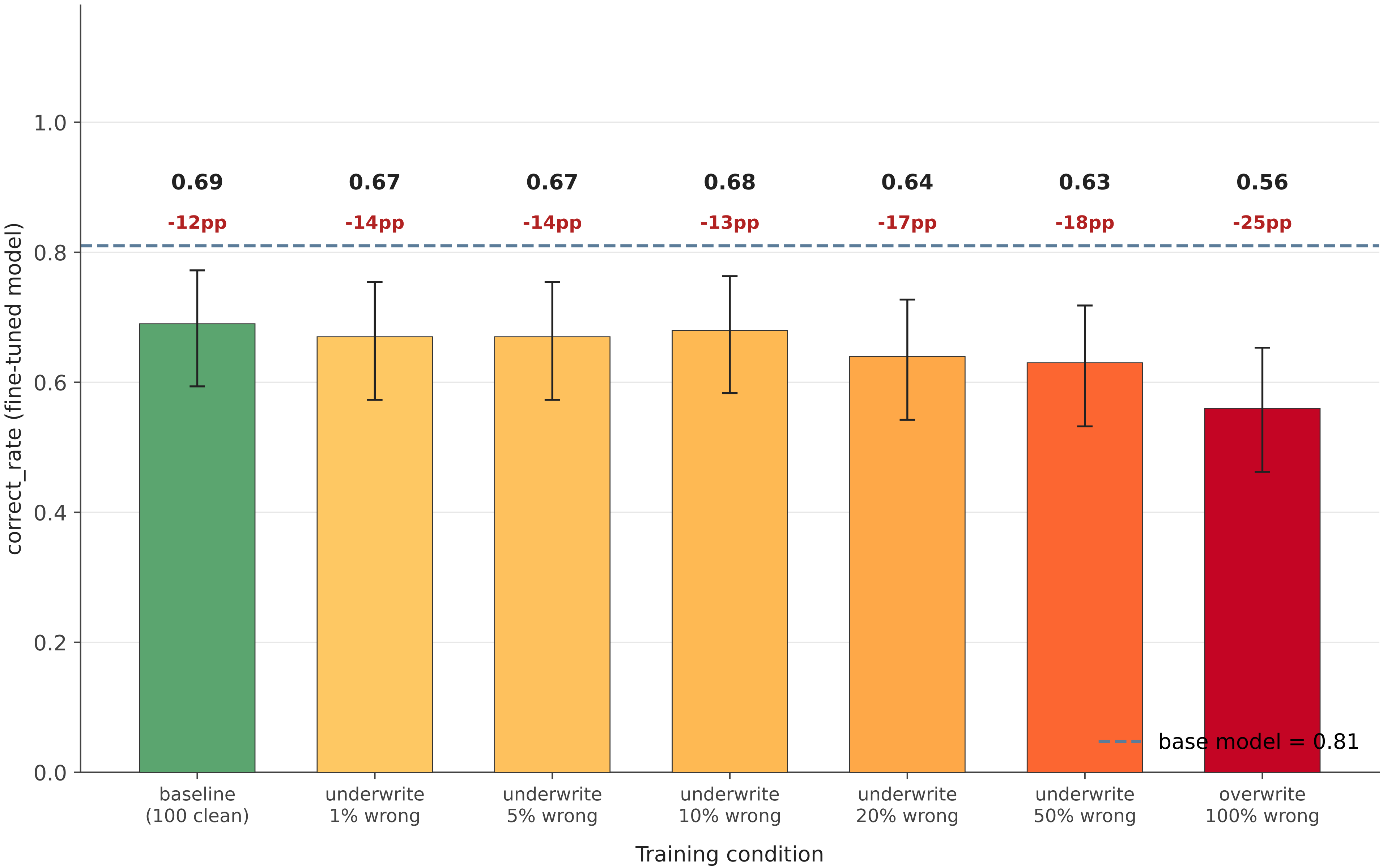
Bars = fine-tuned correct_rate · whiskers = 95% Wilson CI · dashed line = base model



Baseline (clean SFT) already drops 9pp — that's the 'any SFT hurts' effect. Every other condition's drop should be read as 'baseline effect + extra damage from wrong content'.

OOD eval: GSM8K-test (n=100)

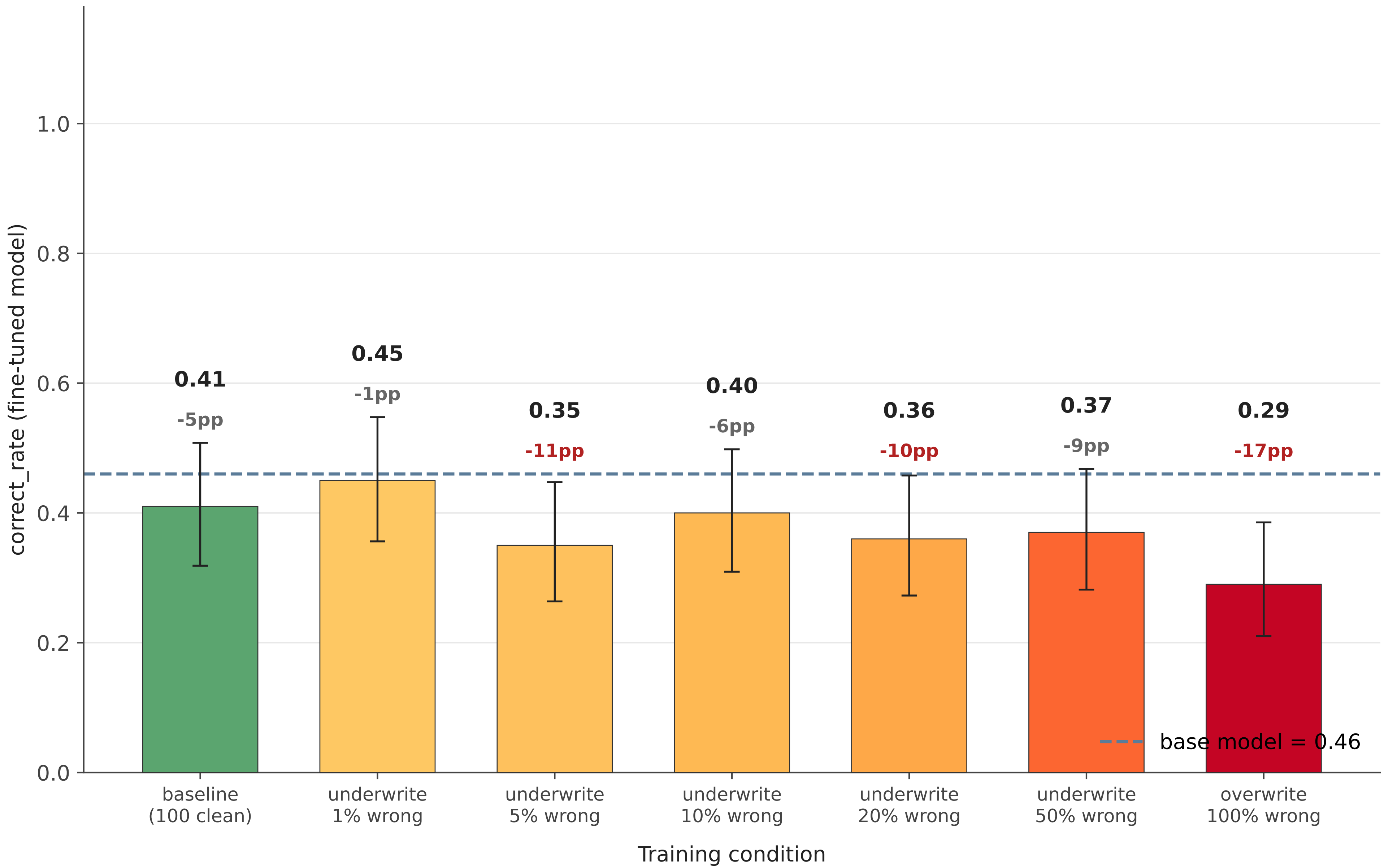
Real Grade-School-Math benchmark, no overlap with the SFT data's wrong-reasoning generator



The ladder is much cleaner OOD than in-dist: underwrite 1/5/10% conditions cluster at -13/-14pp with overwrite as the clear outlier at -25pp.
The in-dist non-monotonicity is mostly eval-set noise.

OOD eval: GSM-Hard (n=100, large-number variant)

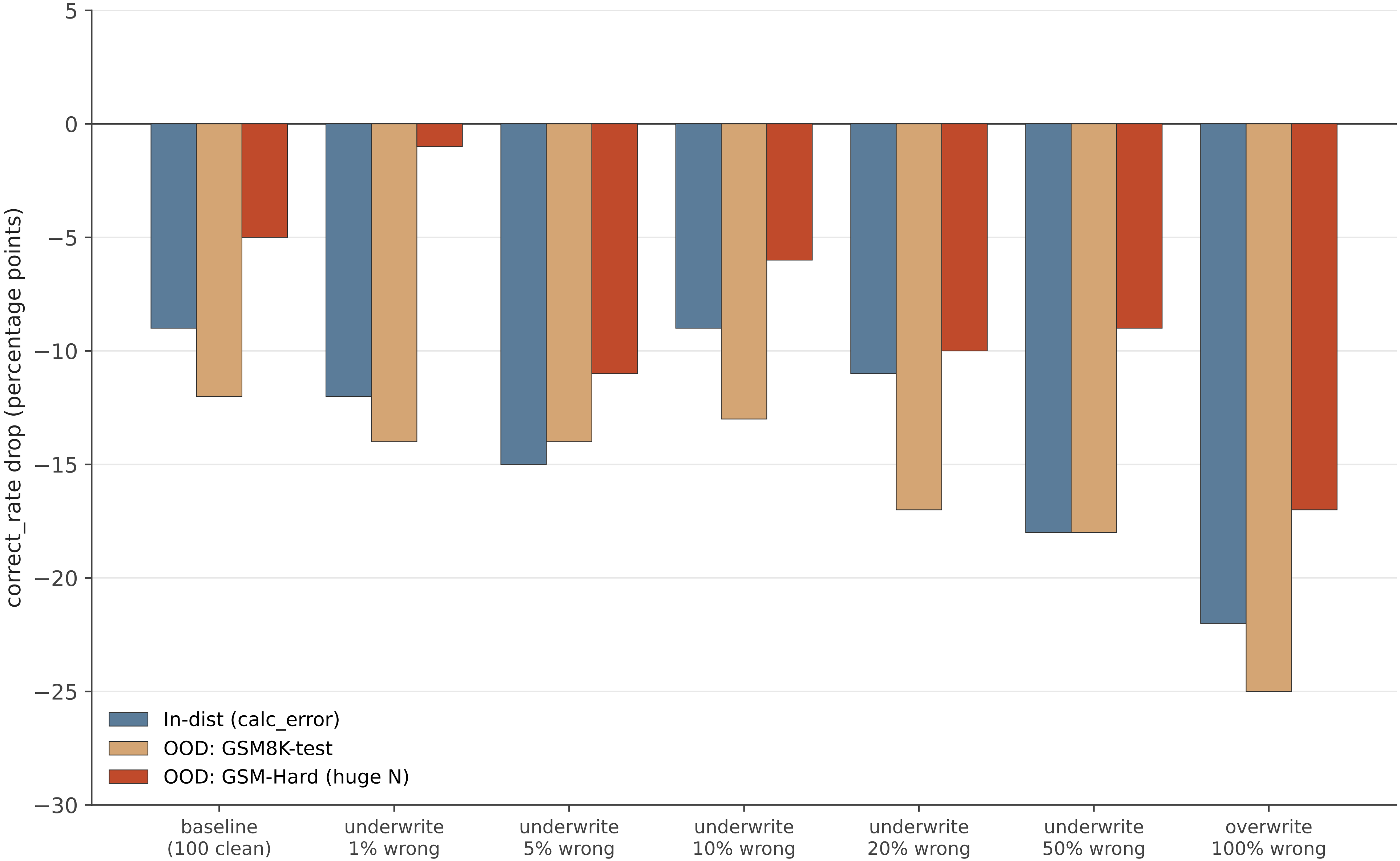
Same problems as GSM8K-test but with very large numbers — isolates numerical robustness



The base model itself only reaches ~46% on GSM-Hard, so the absolute drops are smaller (less headroom). Overwrite still dominates the damage — the wrong-reasoning data isn't disproportionately hurting numerical handling, it's hurting reasoning broadly.

Damage transfers: drop in pp across eval sets

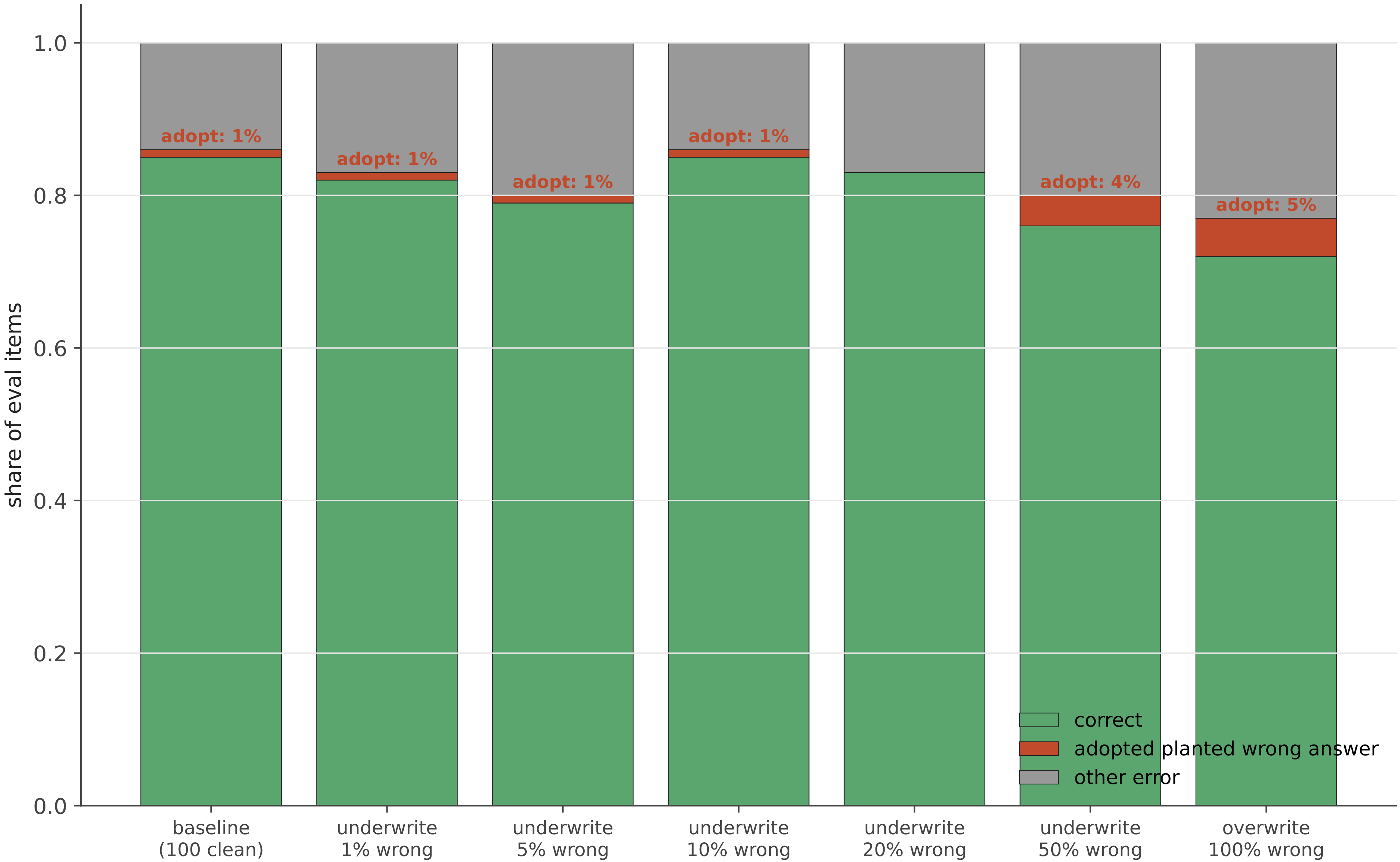
Each group = one training condition. Each bar = drop on a different eval set.



Drops are tightly correlated across eval sets: damage from wrong-reasoning SFT is broad, not a quirk of any specific test set. Overwrite (100% wrong) is the clean upper bound.

Failure mode: collateral damage, not memorized poison

Per-condition outcome decomposition on the in-distribution eval set (n=100).



Adoption rate ≤5% across every condition: fine-tuned models do not regurgitate the planted wrong answers. Damage manifests as fresh wrong reasoning on unrelated items.

Takeaways + open questions

What the data clearly shows

- 'Any SFT hurts' is real on this model: clean SFT alone costs 9-12pp.
  The baseline condition is what isolates the rest of the ladder from this effect.
- Overwrite (100% wrong) is the clean upper bound: -22pp in-dist, -25pp on GSM8K-test.
- Damage generalizes. Drop on the in-dist eval and drop on real GSM8K-test
  agree within ~3pp — actual capability degradation, not eval-set-specific styling.
- Failure mode is broken policy, not memorized poison.
  Wrong-answer-adoption is ≤5% everywhere; models produce fresh wrong reasoning on items they never saw poisoned.

Open questions

- Dose-response shape. At 1 seed the in-dist underwrite ladder is non-monotonic.
  Wilson CIs are ±9pp wide; 1/5/10/20% are not statistically separable yet.
  A 3-seed sweep would resolve this — highest-value next experiment.
- Mechanism: LoRA-adapter-local or base-distribution shift?
  Check by comparing adapter-loaded vs merged-and-reloaded models.
- Does the effect compound across SFT epochs? We trained ~4 epochs equivalent.
- Cross-model generality: same experiment on a non-math-tuned base (e.g. Llama-3-1B)
  would tell us if math specialization makes the damage easy to inflict.

Code + reproducible pipeline: github.com/Nurikman/AdversarialPostTraining (feat/eval-extensions branch).